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January 27, 1969

INTERIM FEDERAL SPECIFICATION

RACKS, STORAGE, LOADED PALLET

This Interim Federal Specification was developed by Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use this document in procurement and forward any recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers boltless, single and multiple opening, steel pallet racks.

1.2 Classification. The storage racks shall be one Type, and of the following Styles and Sizes, as specified:

Style A - 54 inch width opening for one 40 inch by 48 inch pallet. Rated shelf capacity of 3500 pounds.

Style B - 108 inch width opening for two 40 inch by 48 inch pallets, side by side. Rated shelf capacity of 6000 pounds.

Size 1 - 120 inches high.
Size 2 - 138 inches high.
Size 3 - 144 inches high.
Size 4 - 168 inches high.

2. APPLICABLE DOCUMENTS.

2.1 Specifications and Standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

NN-P-71 - Pallets, Material Handling, Wood, Double Faced, Stringer Construction.
NN-P-0073 - Pallets, Material Handling, Plywood, Double Faced, Stringer and Block Construction.
TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

Federal Standard :

Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard :

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 7125

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Institute of Steel Construction Publication:

Specification for the Design, Fabrication and Erection of Structural Steel for Buildings

(Applications for copies may be addressed to the American Institute of Steel Construction, 101 Park Avenue, New York, New York 10017.)

American Iron and Steel Institute Publication:

Light Gage Cold-Formed Steel Design Manual

(Application for copies may be addressed to the American Iron and Steel Institute, 150 East Forty-Second Street, New York, New York 10017.)

American Welding Society (AWS) Publication:

A3.0-61 - Definitions, Welding and Cutting

(Application for copies may be addressed to American Welding Society, 33 West 39th Street, New York, New York 10018.)

3. REQUIREMENTS.

3.1 Preproduction sample. Unless otherwise specified (see 6.2), before production is commenced, a sample of the finished commodity and each component part shall be made available for the contracting officer or his authorized representative(s) to examine and test to determine compliance with this specification. Approval of the preproduction sample authorizes the commencement of production, but does not relieve the contractor of the responsibility of complying with all other applicable provisions of this specification. Production units shall not vary from the approved preproduction sample in design or construction without written approval of the contracting officer.

3.2 Materials. All materials used shall be as hereinafter specified. All materials shall be unused and free of all imperfections that affect appearance or serviceability of the finished commodity. Material not specified herein for any component(s) may be a manufacturer's commercial material which will assure serviceability of the component(s). All materials shall meet the applicable test requirements specified in Section 4.

3.2.1 Steel. The steel used for the fabrication of the storage racks shall be in accordance with one of the American Society for Testing and Material (ASTM) designations for structural steel referenced in the Specification for the Design, Fabrication and Erection of Structural Steel for Buildings, or one of the suitable ASTM designations for structural quality sheet or strip referenced in the Specification for the Design of Light Gage Cold-Formed Steel Structural Members. The steel shall be of an ASTM designation grade (when applicable), and of sufficient mass to assure compliance with all test requirements of this specification.

3.2.2 Finishing Materials.

3.2.2.1 Primer. The primer, when used, shall be compatible with the finishing enamel.

3.2.2.2 Enamel. The enamel shall be of air-drying or baking type. A 20 gage steel panel prepared in accordance with the requirements of 3.5 and the manufacturer's finishing schedule for the racks shall pass the following tests:

Hardness: The dried film shall withstand the firm stroke of a 2H pencil held at a 45 degree angle and pushed across the film surface without evidence of marring when viewed at an oblique angle in a strong light.

Flexibility: The dried film shall show no evidence of cracking or flaking under seven power magnification after a finished panel has been bent through 180 degrees over a 1/8 inch mandrel.

Adhesion: The dried film shall not be removed from the panel when the latter has been scored with a razor blade through the film to the base metal in such a manner as to produce a grid of 1/8 inch squares and a one inch wide piece of cellophane tape (Scotch Brand No. 600) is applied firmly to the grid surface and then quickly pulled from the surface.

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3.2.2.3 Color of finish. Unless otherwise specified (see 6.2), the color of the finish shall be gray, color number 26492 of Fed. Std. No. 595. Unless otherwise specified, the color of the finish of the beam-to-post safety lock shall be the color of the rack, or cadmium, chromium, or zinc plated. Sample color chips of the standard colors are obtainable from the Business Service Center, Federal Supply Service, General Services Administration, Washington, D C. 20407, or from the Business Service Center of the nearest GSA Regional office.

3.3 Design. The design of the storage racks shall be similar to the illustrations in Figure 1. The design of all steel load-bearing components, hot-rolled to shape, shall be in accordance with the Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings, as published in the Steel Construction Manual of the American Institute of Steel Construction, except as specified herein. The design of all steel load bearing components, cold-formed to shape, shall be in accordance with the Specification for the Design of Light Gage Cold-Formed Steel Structural Members, as published in the Light Gage Cold-Formed Steel Design Manual of the American Iron and Steel Institute, except as herein specified. The design and construction of each load bearing component shall assure that the component is capable of sustaining the minimum safe load, hereinafter specified, with a safety factor of not less than 1.65.

3.4 Construction. The construction of the storage racks shall be as specified in the subsequent paragraphs. The upright frame, shelf unit, shelf beam, front-to-rear supports and other components specified shall be constructed to produce a rigid, stable rack that will withstand all tests specified in Section 4. The racks shall be capable of being assembled side by side, in a row, and connected back to back in a double row.

3.4.1 Welding. All welding shall be accomplished by a suitable arc welding process referenced in AWS A3.0-61. All welds shall be neatly accomplished, free of holes, poor fusion, slag, flash and spatter. All welded joints shall withstand the test specified in Section 4.

3.4.2 Upright frame. The upright frame shall be of the material specified in 3.2.1, and shall be 40 inches wide. It shall be all welded construction consisting of two posts, diagonal and horizontal braces, and floor bearing plates. Each upright frame shall have a minimum safe load capacity of 24,000 pounds. The post shall be a channel, box channel, rectangular or square formation. The width of the post, facing the aisle, shall be not more than 4-1/2 inches. The bottom of each post shall have a floor bearing plate securely attached, which shall have not less than 13.5 square inches of bearing surface, and shall be of not less than 11 gage (0.1196 inch steel. The plate shall have an accessible 3/4 inch, plus 1/64 inch, minus 0 inch, diameter hole for for anchoring the rack to the floor. Each post shall have openings to readily accommodate two shelf beam connectors, end to end, at corresponding heights throughout the height range of the post openings. A single or double row of openings shall commence not farther than 24 inches from the bottom of the floor bearing plate and continue upward to the top of the post by either 1, 2, 3, or 4 inch increments. All openings of each post shall be identical to each other and identical to the openings of all other posts. The positioning of all corresponding openings of the posts shall not vary more than 3/32 inch as measured from the bottom of the floor bearing plates. The arrangement of the openings and the fastening devices of the shelf unit shall be such that it can be attached so that the top of the shelf unit is flush, plus or minus 3/8 inch, with the top of the post, and 24 inches or less from the bottom of the post. The posts shall be braced by horizontal and diagonal or "X" braces as illustrated in Figure 1. The braces shall be an angle, channel, box channel, rectangular, square, or "Z" formation. A minimum of four horizontal and three diagonal or "Z" braces shall be required for upright frames up to and including 144 inches in height. A minimum of five horizontal and four diagonal or "X" braces shall be required for upright frames over 144 inches in height. Each end of all braces shall be securely welded to the posts, and the "X" brace members may be welded to each other, at the point of intersection. The braces shall not protrude beyond the sides of the posts or obstruct the attachment of the shelf beams to the post, at any height on the posts. The center line of the bottom horizontal brace shall be located 6 inches plus or minus 1/4 inch, from the bottom of the upright frame, and the center line of the top horizontal brace shall be located not farther than 6 inches from the top of the upright frame. Other horizontal braces shall be uniformly spaced between the top and bottom horizontal braces. The posts shall be parallel, and the horizontal braces shall be parallel to each other and square to the post. The upright frames of Style A and Style B racks, supplied on any one contract, shall be interchangeable. The upright frames shall conform to the tests and requirements specified in 4.4.2.

3.4.3 Shelf unit. The shelf unit for Style A racks shall have a minimum safe load capacity of 3500 pounds, and shall consist of an integral welded unit of two, 54 inch length beams and two front-to-rear supports. The shelf unit for Style B racks shall have a minimum safe load capacity of 6000 pounds, and shall consist of an integral welded unit of two, 108 inch length beams and four front-to-rear supports. Alternatively, the shelf unit of Styles A and B racks may consist of two beams of proper length and the respective number of drop-in or slip-in type front-to-rear supports. The shelf units shall be similar to the illustrations in Figure 1. The height of the assembled shelf unit shall not exceed 5 inches and the depth (front-to-rear) shall be compatible with the width of the upright frame. All shelf units shall conform to the tests and requirements specified in 4.4.4 and 4.4.5.

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3.4.3.1 Shelf beam. The shelf beam shall be of one or two piece construction of the material specified in 3.2.1. The one-piece beam formation shall have not less than two, 90 degree bends, and the two-piece beam formation shall have not less than four 90 degree bends. The beam formation(s) shall be formed accurately and uniformly, and the pieces of the two-piece beam shall be joined neatly and firmly on top and bottom, by chain or staggered intermittent welds, spaced on centers not farther than 12 inches, as measured on the line of the welds. When drop-in or slip-in front-to-rear supports are used, the beams shall have openings so as to locate the front-to-rear supports as illustrated in Figure 1. Each end of the beams shall have a multiple prong slip-fit or slip-joint beam-to-post connector in accordance with, or similar to, the illustration in Figure 1. The connector shall be capable of transferring all loads and external forces of the shelf beam to the post without failure of the connection, or damage to the post or connector. The prongs shall be of sufficient number, size, and strength to withstand repeated loading and unloading of the rack, and to continually sustain the rated shelf load. The prongs shall engage with the openings of the post in such a manner as to prevent lateral rotation of the shelf beam, with each prong contributing to the support of the shelf unit. The beam shall conform to the tests and requirements specified in 4.4.4 and 4.4.5.

3.4.3.2 Beam-to-post safety lock. Each beam-to-post connection shall have a safety lock which will lock the beam to the post. The design and manner of construction of the device shall prevent accidental disengagement of the beam-to-post connection during movement or shifting of the assembled rack, and impact to the upright frame or shelf unit from horizontal or vertical directions. The safety lock may be an integral part of the shelf beam or may consist of additional components. The safety lock shall be employed without the use of tools, and no part thereof shall obstruct or restrict the loading or unloading of the shelf units.

3.4.3.3 Front-to-rear supports. The front-to-rear supports shall be of the material specified in 3.2.1. The supports shall be a channel, box channel, hatsection, rectangular or square formation, with a minimum width of 2 inches. Each front-to-rear support shall have a minimum safe load capacity of 1500 pounds. The supports shall locate on the shelf beams as illustrated in Figure 1, and the top of the supports shall be flush, plus 0 inch minus 1/16 inch, with the top of the shelf beam. Slip-in or drop-in supports shall attach to the shelf beams in such a manner as to prevent: lateral movement of the supports when laterally hit with a soft hammer; the releasing and dropping of the support when vertically hit on the bottom of the support with a soft hammer, and the releasing and dropping of the support upon skewing of the beams(s) when subjected to the capacity or test loads. The supports shall attach and detach without the use of tools, other than a soft hammer. Supports with hanger-type attachment may have hangers that extend beyond the sides of the shelf beams, but not more than 3/16 inch. The front-to-rear supports shall conform to the tests and requirements specified in 4.4.3.

3.4.4 Back-to-back connector. When specified (see 6.2) back-to-back connector(s) shall be furnished with the rack unit. The back-to-back connectors shall be capable of joining the backs of the same Style units together to form a double row arrangement. The connectors may be the fixed or adjustable type, but shall provide 4-inch spacing between the connected posts. Each end of the connector shall firmly and securely attach to any post of the upright frames, at not less than 2 locations within the lower and upper limits of the shelf unit attachment, by bolts, washers, and nuts. Each end shall attach to the post at not less than 2 points of attachment. Alternatively, the attachment may be boltless, with not less than 2 points of attachment at each post, at not less than 2 locations within the lower and upper limits of the shelf unit attachment. The boltless attachment shall have a safety locking device to prevent unintentional disengagement. The back-to-back connector shall be of sufficient size and strength to withstand all forces applied to the connected rack units, during normal loading and unloading operations, without deforming or failure.

3.4.5 Leveling shim. When specified (see 6.2), leveling shims shall be provided with the rack units. The shims shall be hot or cold rolled steel with a minimum thickness of 12 gage (0.1046 inch), and of same size as the floor bearing plate. The shims shall have a hole corresponding to the hole in the floor bearing plate, and shall anchor or lock with the plate when the unit is erected.

3.4.6 Dimensions. The dimensions of the racks and components shall be as specified herein, and as shown in Figure 1. See 1.2 for height and width of the racks.

3.4.6.1 Tolerances. Unless otherwise specified, the following tolerances shall apply:

Any component dimension - plus or minus 1/32 inch.
Overall depth, height and width - plus or minus 3/32 inch.

The tolerances referenced above may be cumulative but not to the extent that critical, specified, or overall tolerances cannot be maintained, or the components and sub-assemblies will not easily assemble.

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3.4.7 Assembly. The upright frames and the shelf units shall assemble and disassemble, throughout the height range of the post openings, easily and freely, without the use of bolts or tools other than a soft hammer. Adjacent shelf units shall assemble and disassemble with the common upright frame(s), easily and freely, without the use of tools other than a soft hammer, at corresponding heights throughout the height range of the post openings. An assembled rack of two upright frames, and three shelf units shall be rigid, stable and without deformation or skewing when each shelf unit is loaded with the capacity or test loads. The rack shall assemble so that the upright frames are perpendicular, parallel to each other, and reasonably square with the shelf units, and the shelf units are parallel to each other. The upright frames of Styles A and B racks, provided on any one contract, shall assemble with either end of all shelf units of Styles A and B racks. The drop-in or slip-in front-to-rear supports shall assemble and disassemble with the shelf beams, easily and freely, without the use of bolts or tools other than a soft hammer. The assembly of the front-to-rear supports to the shelf beams shall prevent the releasing and dropping of the supports as specified in 3.4.3.3. The beam-to-post safety lock shall engage and disengage without the use of tools, and shall be effective at any position within the range of the shelf unit attachment. The leveling shim shall anchor or lock with the floor bearing plate in such a manner as not to separate during normal use of the rack.

3.5 Pretreatment and finishing.

3.5.1 Pretreatment. All surfaces shall be treated for finishing in accordance with any method of TT-C-490. The pretreatment surfaces shall be free of scale, oil, grease, rust, wax, and foreign matter and deposits deleterious to the finish.

3.5.2 Priming. The surface prepared for finishing shall have a coating of primer specified in 3.2.2.1, or the surfaces shall be prepared for finishing in accordance with Type I or Type II or TT-C-490.

3.5.3 Enameling. All surfaces prepared for finishing as specified in 3.5.1 and 3.5.2 shall be coated with the enamel specified in 3.2.2.2. The total dry film thickness of the primer and enamel shall be not less than 1.1 mils on all exposed (exterior) surfaces and not less than 0.7 mils on all other surfaces. The dry film thickness of the enamel on exposed (exterior) surfaces prepared for finishing in accordance with Type I or Type II of TT-C-490 shall be not less than 0.8 mils and not less than 0.7 mil on all other surfaces. The color of the finish shall correspond to the color number referenced in 3.2.2.3. The finish shall level out to produce a well bonded, smooth, hard surface, free of runs, foreign matter, area of thin or no film, and separation of color.

3.6 Identification marking. Unless otherwise specified (see 6.2), the components of the rack shall be free of identification marking.

3.7 Workmanship. The workmanship exercises while performing the various operations of manufacturing shall produce a serviceable product acceptable under the requirements specified herein. All welds, edges, and corners shall be smooth, free of sharp projections and burrs. Formed parts shall be uniform, without cracks, fractures, and undesirable tears and holes. The openings and holes shall be accurate and uniform. All referenced dimensions shall be within specified tolerance. All movable parts shall operate freely and function properly. Attention shall be given to pretreatment operations to assure that all surfaces are pretreated and the proper sequence of operation is maintained and completed. The finish shall be applied in accordance with good commercial practice to assure complete coverage, specified dry film thickness, and durability of the finish. The degree of workmanship shall assure that components will assemble and disassemble as specified in 3.4.7, and the occurence of defects will not exceed the acceptable quality level specified herein.

3.8 Conflict of text and figure. In any case of conflict between the text and figure 1, the text shall prevail.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any or all of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and Material Inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.2 Sampling for inspection and acceptance. Unless otherwise indicated, sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.2.1 End item inspection.

4.2.1.1 Visual examination. The examination of the racks shall be in accordance with the classification of defects specified in Table I, and the inspection and acceptable quality levels (AQL's) specified in Table III. The lot size shall be expressed in units of racks for the purpose of determining the sample size in accordance with MIL-STD-105. The sample unit for this examination shall be one complete rack, consisting of two end frames and three shelf units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Steel	Steel previously used or not new stock.	X	
	Steel contains scale, rust, pits, fins, sharp edges, protrusions, burrs, or undesirable holes.	X	
Welds	Welding method employed other than specified.		X
	Welds not neatly accomplished, burn thru, cracked, or poorly fused.	X	
	Rough, sharp projections, flash and spatter not removed.		
Pretreatment and finish	Pretreatment omitted, primer omitted when required, and area(s) of no film.	X	
	Poor adhesion, blistering, or peeling.	X	
	Finish contains runs, sag, area(s) of thin film, or separation of color.		X
	Color of finish other than specified.		X
	Thickness not as specified.		X
Upright frames	Frame design not similar to illustration in Figure 1, or not constructed as specified.	X	
	Number of horizontal and diagonal braces less than minimum specified.	X	
	Top and bottom horizontal braces not located as specified.		X
	Floor bearing plate not attached, or size is less than minimum specified.	X	
	Hole missing, wrong size, or not accessible in floor bearing plate.		X
	Openings in posts will not accommodate two shelf beam connectors end to end.	X	
	Openings in posts missing or not spaced in increments specified.	X	
	Leveling shim will not anchor or lock with floor bearing plate.		X
Shelf units	Not similar to illustration in Figure 1.	X	
	Assembled height exceeds 5 inches.		X
	Shelf beam formation not uniform.		X
	Safety lock or locks missing, ineffective or inoperative.	X	
	Front-to-rear supports not located or not capable of being attached to shelf beams as shown in Figure 1.	X	
	Front-to-rear supports (drop-in or slip-in type) will not firmly attach to shelf beams.	X	

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TABLE I. Classification of Defects (con.)

Examine	Defect	Classification Major	Minor
Back-to-back connector	Incapable of providing 4-inch spacing between posts.	X	
	Each end has less than 2 points of attachment to post.		X
	Boltless attachment not equipped with safety locking device.	X	
Assembly	Components will not assemble and disassemble without the use of tools other than a soft hammer.		X
	Shelf units will not assemble with posts throughout the range of openings.	X	
	Style A rack components will not assemble components of Style B racks.	X	
	The shelf unit will not assemble with upright frames within top and bottom limits specified in 3.4.2.		X
	Assembled rack is not rigid and stable when loaded with capacity or test loads.	X	
Workmanship	Missing part or component, except bolt, washer or nut.	X	
	Missing bolt, washer, or nut.		X
	Malformed or damaged part.	X	
	Burrs, sharp edges or sharp projections on welds, corners, or edges.	X	
	Undesirable holes or tears in parts.		X
	Holes and openings not accurate or uniform.		X

4.2.1.2 Dimensional examination. Examination will be made of the racks for compliance with dimensions and tolerances specified. Any dimension not within tolerance specified shall be classified as a defect. The inspection and acceptable quality level shall be as shown in Table III.

4.2.2 Examination of preparation for delivery. An examination will be made to determine that packaging, packing, and markings are in accordance with Section 5 of this specification. Defects will be enumerated in accordance with Table II below. The sample unit will be one shipping container, and the lot size will be the number of containers in the inspection lot. The inspection and acceptable quality level shall be as shown in Table III.

TABLE II. Classification of preparation for delivery defects

Examine	Defect
Marking	Missing, incorrect, illegible, improper size, location, sequence, or method of application.
Material	Other than specified, damaged or otherwise defective affecting serviceability. Missing damaged, or nonconforming component(s).
Workmanship	Inadequate application of material, loose components, improper separation and padding or components.

4.3 Inspection levels and acceptable quality levels. The inspection levels and the acceptable quality levels expressed in defects per hundred units shall be in accordance with MIL-STD-105, and as shown in Table III.

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TABLE III - Inspection and Acceptable Quality Levels

	Inspection Level	Acceptable Quality Level Major	Total
For defects listed in 4.2.1.1	S-2	4.0	10.0
For defects listed in 4.2.1.2	S-2	one class	4.0
For defects listed in 4.2.2	S-2	one class	4.0

4.4 Tests.

4.4.1 Load test. The following load tests shall be performed using inspection level S-1 for the selection of the number of racks to be tested from any lot. The lot size shall be expressed in units of racks. All sizes of Styles A and B racks shall be subject to the load test. Each unit tested shall be a completely assembled rack of two uprights and three shelf units. The shelf units may be attached to the upright frames at convenient heights, except the top shelf unit shall be located where it will be free of any load applied to the top of the upright frames. All loads referenced in the individual test shall have a tolerance to plus or minus one-half of one percent however, the minimum allowable load(s) shall not be intentionally used for the tests. All tests shall be conducted in the presence of the contracting officer or his authorized representative(s). The government reserves the right to use additional equipment, information or methods deemed necessary to determine compliance with the requirements of this specification. The sequence of the test may be as herein delineated or any sequence of test favorable to existing circumstances may be used. The contractor has the option of having all tests performed on the racks as initially set up for testing, or he may substitute new components for those that have passed their particular test, before performing the next test. In the event the substituted components fail while subsequent tests are being performed, the previously recorded satisfactory performance for those components shall be declared void. Test loads mechanically applied to the pallets may be utilized provided this method does not preclude compliance with any other requirement in the tests specified herein.

4.4.1.1 Upright frames. A load of 24,000 pounds shall be applied to the two posts of each upright frame. The load shall be applied axially and as evenly as possible to the posts. The load shall remain on the posts for a period of one hour. At the end of one hour, the upright frames shall be visually examined, with the load applied, for evidence of deformation, skewing, joint separation, metal rupture, or component failure. Upon evidence of any of the above stated defects, the test shall be stopped and the rack unit shall be considered as failing to meet the requirements of this specification. In absence of any of the above defects, the load shall be increased to 39,600 pounds. This load shall be applied for 30 minutes and then the upright frames shall be re-examined, with the load applied, for evidence of the defects stated above. Evidence of any of the defects stated above shall be sufficient cause for rejection of the rack unit.

4.4.1.2 Front-to-rear supports. A load of 3,000 pounds shall be applied to each assembled pair of front-to-rear supports of one shelf unit. The load shall be distributed as uniformly as possible over the length of the supports that is between the shelf beams. Each pair of front-to-rear supports of Style B racks may be tested separately or simultaneously. The load shall be applied for one hour. At the end of one hour, the load shall be removed and the supports shall be visually examined for evidence of deformation, skewing, joint separation, metal rupture, or component failure. Upon evidence of any of the defects stated above, the test shall be stopped, and the rack unit shall be considered as failing to meet the requirements of this specification. In absence of any of the defects stated above, the load shall be increased to 4,950 pounds. The load shall be applied for 30 minutes, and then removed. The supports shall be re-examined for evidence of the defects stated above. Evidence of any of the defects stated above shall be sufficient cause for rejection of the rack unit.

4.4.1.3 Shelf unit - Style A racks. A 3500 pound load shall be distributed as uniformly as feasible on a 40 inch by 48 inch wood pallet placed on each shelf unit. The shelf units shall have the loads simultaneously for a period of one hour. The deflection of any shelf beam, under load, in excess of 0.200 inch shall be sufficient cause for rejection of the rack unit. The measurement shall be made with a measuring device or devices with a unit calibration which does not exceed 0.010 inch. The measurement may be accomplished by a dial indicator, with an acceptable calibration, and precision space blocks, working from precision ground reference surface(s). Upon evidence of shelf beam deflection in excess of 0.200 inch, joint separation, metal rupture, or component failure the test shall be stopped and the rack unit shall be considered as failing to meet the requirements of this specification. In absence of any of the defects stated above, the load on

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each pallet shall be increased to 5,775 pounds. This load shall be applied for 30 minutes and then removed. The unloaded rack unit shall be visually and dimensionally examined for evidence of permanent deformation, skewing, joint separation, metal rupture and component failure. Evidence of any of the defects stated above shall be sufficient cause for rejection of the rack unit.

4.4.1.4 Shelf unit - Style B racks. A 6,000 pound load shall be equally divided and placed on two 40 inch by 48 inch wood pallets, and the load on each pallet shall be distributed as uniformly as feasible. Two such loaded pallets shall be placed on each shelf unit, and the shelf units shall have the loads applied simultaneously for a period of one hour. The deflection of any shelf beam, under load, in excess of 0.400 inch shall be sufficient cause for rejection of the rack unit. The measurement shall be as specified in 4.4.1.3. Upon evidence of shelf beam deflection in excess of 0.400 inch, joint separation, metal rupture, or component failure, the test shall be stopped and the rack unit shall be considered as failing to meet the requirements of this specification. In absence of any of the defects stated above, the load on each pallet shall be increased to 4,950 pounds. This load shall be applied for 30 minutes and then removed. The unloaded rack unit shall be visually and dimensionally examined for evidence of permanent deformation, skewing, joint separation, metal rupture and component failure. Evidence of any of the defects stated above shall be sufficient cause for rejection of the rack unit.

4.4.1.5 Pallets for test loads. The pallets used when performing the tests specified in 4.4.1.3 and 4.4.1.4 shall be of the size specified therein and shall be of any construction covered by NN-P-71 or NN-P-0073.

5. PREPARATION FOR DELIVERY.

5.1 Packing. Unless otherwise specified, like components of the racks shall have the finished surfaces protected by corrugated fiberboard seapartors. The seapartors shall be located between each layer of components. For handling by forklift or sling, 3-inch by 4-inch wooden runners shall be provided. The entire stack shall be banded with steel strapping commonly used for the purpose. Staples shall be used to position the strapping to the runners.

5.2 Marking. Unless otherwise specified, the strapped components shall be marked with the name of the components material, style, size and quantity contained therein as defined by the contract or order under which shipment is made, the name of the contractor, number of the contract or order and the gross weight.

6. NOTES.

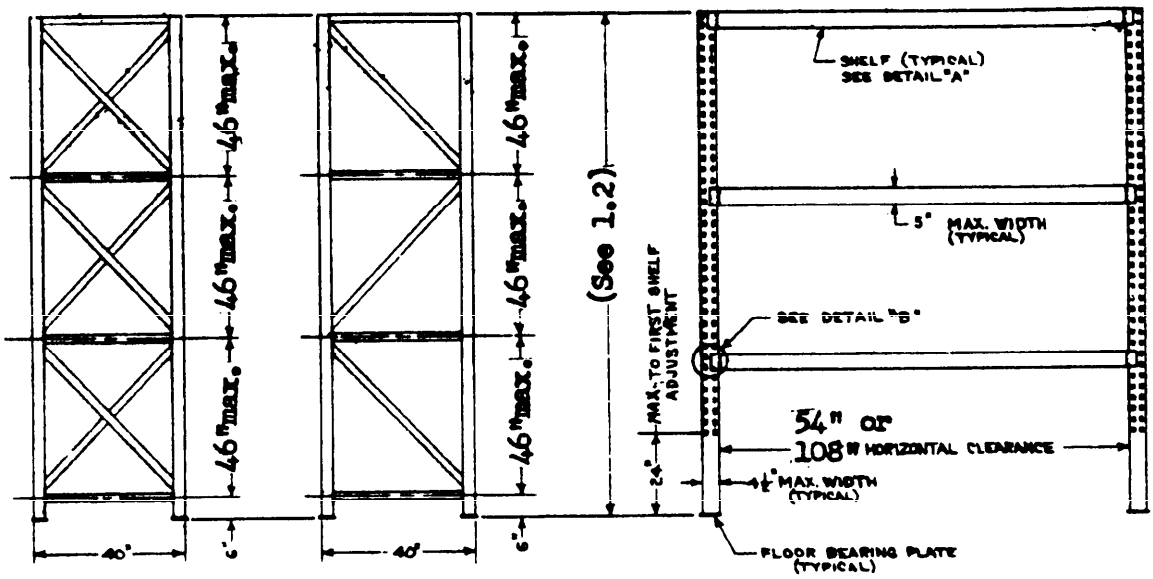
6.1 Intended use. The boltless racks covered by this specification are intended for storing palletized loads, in such a manner to permit placement and removal of any individual pallet without disturbing any other pallet(s).

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Style and Size rack units desired (see 1.2).
(c) Whether preproduction sample is not required (see 3.1).
(d) Whether color, other than No. 26492 of Fed. Std. No. 595, is required (see 3.2.2.3).
(e) Whether color, other than the color of the rack, cadmium, chromium, or zinc is required for the safety lock (see 3.2.2.3).
(f) Whether back-to-back connectors shall be furnished (see 3.4.4).
(g) Whether leveling shims shall be furnished (see 3.4.5).
(h) Whether identification marking is required (see 3.6).

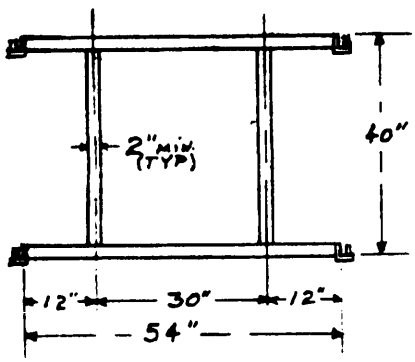
6.3 Miscellaneous. This specification does not include all types, styles, and sizes of rack units manufactured. The boltless rack units covered by this specification are of style and size that meet the present requirements of the Federal Government.

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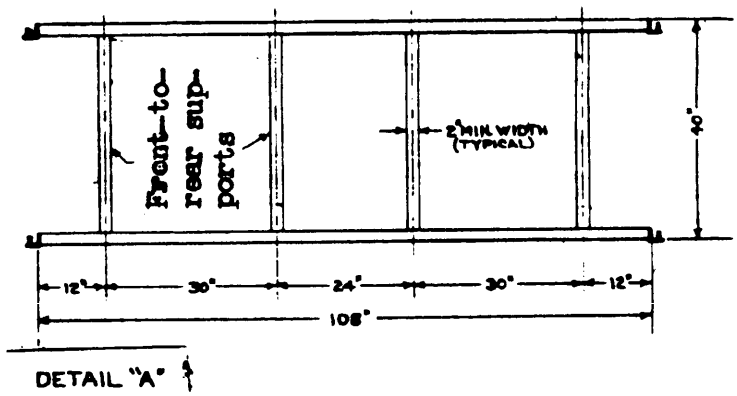


OPTIONAL UPRIGHT FRAME CONSTRUCTION

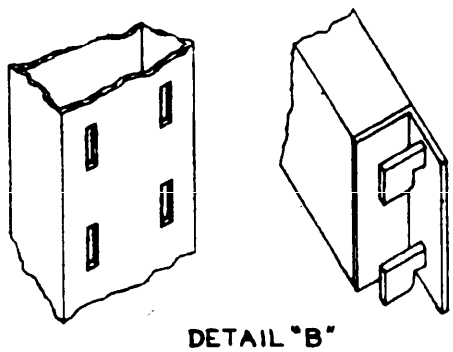
STORAGE RACK ASSEMBLY



TYPICAL STYLE A RACK SHELF UNIT



TYPICAL STYLE B RACK SHELF UNIT



TYPICAL BEAM-TO-POST CONNECTOR

FIGURE 1. Racks, storage, loaded pallet

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